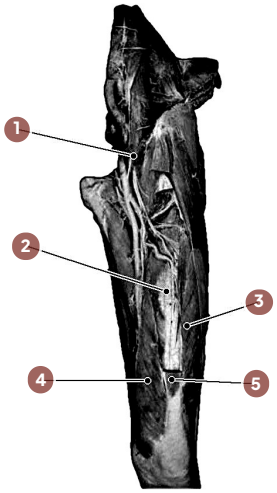
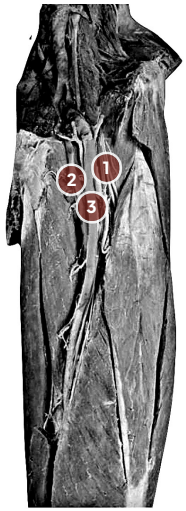
Lower Limb: Muscles, Nerves and Vessels

Competency 127 and the lab muscle list: thigh, gluteal region and leg on the cadaver, with the nerves and vessels that run with them. Name _____

A. Anterior thigh and the femoral triangle (cadaver) COMPETENCY 127, LAB



Cadaver, anterior thigh, deep



Cadaver, femoral triangle

L IN

adriceps femoris = rectus femoris, vastus lateralis, intermedius, medialis

adriceps action: extend the knee

psoas action: flex the hip

ıngle borders: inguinal ligament, sartorius, adductor longus

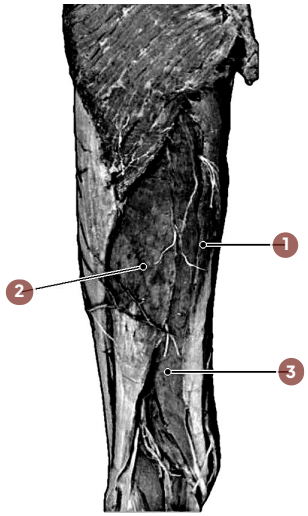
eral to medial: nerve, artery, vein (NAVEL)

ıoral artery was the external iliac artery

1. Iliopsoas
2. Vastus intermedius
3. Vastus lateralis
4. Vastus medialis
5. Rectus femoris

1. Femoral nerve
2. Femoral vein
3. Femoral artery

B. Gluteal region and posterior thigh (cadaver) COMPETENCY 127, LAB



Cadaver, posterior thigh: the hamstrings



Cadaver, deep gluteal region: sciatic nerve

biceps femoris, semitendinosus, semimembranosus

tions: flex the knee, extend the hip

nus: extends the hip

us: abducts the hip, steadies the pelvis

eaves the pelvis greater sciatic foramen, below piriformis

gluteus maximus, hamstrings

he knee into tibial and common fibular

1. Semimembranosus
2. Biceps femoris
3. Semitendinosus

1. Piriformis
2. Gluteus minimus
3. Sciatic nerve

Extra notes and drawings

Extra notes and drawings

Lower Limb: Muscles, Nerves and Vessels

A. Anterior thigh and the femoral triangle (cadaver) · Competency 127, lab · one card per numbered structure, same lines as the slide. Write the name, then fill each line as we go.

Name _____

1 Iliopsoas

Def Psoas major and iliacus acting as one muscle, the main hip flexor
Origin Psoas major: bodies and transverse processes of T12 to L5. Iliacus: iliac fossa
Insertion Lesser trochanter of the femur
Action Flexes the hip; psoas also flexes the trunk
Nerve Psoas: L1 to L3 ventral rami. Iliacus: femoral nerve
Blood supply Lumbar arteries, iliolumbar artery, medial femoral circumflex
Clinical Psoas sign: pain on passive hip extension with an inflamed appendix lying on the muscle

2 Vastus intermedius

Def Deep head of the quadriceps, under rectus femoris
Origin Anterior and lateral shaft of the femur
Insertion Patella and tibial tuberosity by the patellar ligament
Action Extends the knee
Nerve Femoral nerve (L2 to L4)
Blood supply Lateral femoral circumflex artery

3 Vastus lateralis

Def Lateral head of the quadriceps, the largest
Origin Greater trochanter and lateral lip of the linea aspera
Insertion Patella and tibial tuberosity by the patellar ligament
Action Extends the knee
Nerve Femoral nerve (L2 to L4)
Blood supply Lateral femoral circumflex artery
Clinical Common intramuscular injection site in infants

4 Vastus medialis

Def Medial head of the quadriceps, the teardrop above the knee
Origin Intertrochanteric line and medial lip of the linea aspera
Insertion Patella and tibial tuberosity by the patellar ligament
Action Extends the knee; its lowest fibers hold the patella in its groove
Nerve Femoral nerve (L2 to L4)
Blood supply Femoral artery branches
Clinical Weak vastus medialis lets the patella track laterally (patellofemoral pain)

5 Rectus femoris

Def Quadriceps head that crosses both the hip and the knee
Origin Anterior inferior iliac spine
Insertion Patella, then the tibial tuberosity by the patellar ligament
Action Extends the knee, flexes the hip
Nerve Femoral nerve (L2 to L4)
Blood supply Lateral femoral circumflex artery
Clinical Strained in sprinters and kickers; the only quadriceps head that can be torn at the hip

1 Femoral nerve

Def Largest branch of the lumbar plexus
Origin L2 to L4
Location Enters the thigh under the inguinal ligament, lateral to the femoral artery (NAVEL, lateral to medial)
Supplies Quadriceps, sartorius, pectineus, iliacus; skin of the anterior thigh and, by the saphenous nerve, the medial leg
Clinical Injury weakens knee extension and loses the patellar reflex

2 Femoral vein

Def Deep vein of the thigh
Location Medial to the femoral artery in the femoral triangle
Receives Great saphenous vein, deep femoral vein
Course Becomes the external iliac vein above the inguinal ligament
Clinical Site for central venous access; a clot here is a DVT that can reach the lungs

3 Femoral artery

Def Main artery of the thigh
Origin The external iliac artery once it passes under the inguinal ligament
Course Through the femoral triangle between the nerve and the vein; gives the deep femoral artery; passes the adductor hiatus to become the popliteal artery
Supplies Thigh, through its deep femoral branch
Borders Femoral triangle: inguinal ligament above, sartorius lateral, adductor longus medial
Clinical Femoral pulse below the inguinal ligament; the usual entry point for catheters

Lower Limb: Muscles, Nerves and Vessels

B. Gluteal region and posterior thigh (cadaver) · Competency 127, lab · one card per numbered structure, same lines as the slide. Write the name, then fill each line as we go.

Name _____

1 Semimembranosus

Def Deep medial hamstring with a flat membranous tendon
Origin Ischial tuberosity
Insertion Medial condyle of the tibia
Action Flexes the knee, extends the hip, medially rotates the flexed leg
Nerve Tibial division of the sciatic nerve (L5 to S2)
Blood supply Perforating branches of the deep femoral artery
Borders Medial edge of the popliteal fossa

2 Biceps femoris

Def Lateral hamstring with two heads
Origin Long head: ischial tuberosity. Short head: linea aspera
Insertion Head of the fibula
Action Flexes the knee; long head extends the hip
Nerve Long head: tibial division of the sciatic. Short head: common fibular division
Blood supply Perforating branches of the deep femoral artery
Borders Its tendon is the lateral edge of the popliteal fossa
Clinical Hamstring strain at the ischial tuberosity in sprinters; its tendon is the landmark for the common fibular nerve at the knee

3 Semitendinosus

Def Superficial medial hamstring with a long cord-like tendon
Origin Ischial tuberosity
Insertion Medial surface of the upper tibia (pes anserinus)
Action Flexes the knee, extends the hip, medially rotates the flexed leg
Nerve Tibial division of the sciatic nerve (L5 to S2)
Blood supply Perforating branches of the deep femoral artery
Borders With semimembranosus, the medial edge of the popliteal fossa
Clinical Its tendon is harvested for ACL reconstruction

1 Piriformis

Def Deep gluteal muscle and the landmark of the gluteal region
Origin Anterior surface of the sacrum
Insertion Greater trochanter, upper border
Action Laterally rotates the extended hip, abducts the flexed hip
Nerve Nerve to piriformis (S1, S2)
Blood supply Superior and inferior gluteal arteries
Borders Superior gluteal vessels and nerve leave above it; sciatic nerve and inferior gluteal vessels leave below it
Clinical Piriformis syndrome: the muscle compresses the sciatic nerve, giving buttock pain down the leg

2 Gluteus minimus

Def Deepest of the three gluteal muscles
Origin Outer surface of the ilium between the anterior and inferior gluteal lines
Insertion Greater trochanter, anterior surface
Action Abducts and medially rotates the hip; keeps the pelvis level when the opposite foot is off the ground
Nerve Superior gluteal nerve (L4 to S1)
Blood supply Superior gluteal artery
Clinical Weak gluteus medius and minimus give the Trendelenburg sign: pelvis drops on the unsupported side

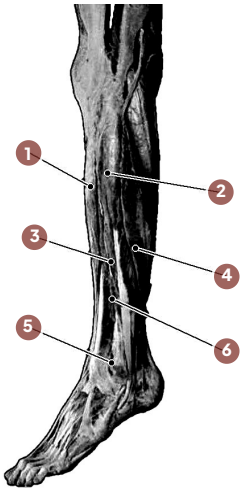
3 Sciatic nerve

Def Largest nerve in the body
Origin Sacral plexus, L4 to S3
Course Leaves through the greater sciatic foramen below piriformis, runs deep to gluteus maximus and the hamstrings, splits above the knee
Branches Tibial nerve and common fibular nerve
Supplies Hamstrings, then the whole leg and foot through its branches
Clinical Sciatica from a herniated disc; an injection in the upper outer quadrant of the buttock avoids it

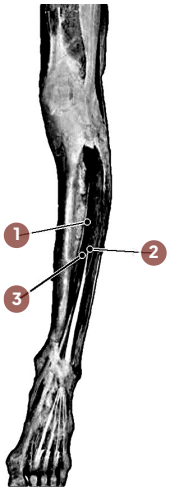
Lower Limb: Muscles, Nerves and Vessels

Competency 127 and the lab muscle list: thigh, gluteal region and leg on the cadaver, with the nerves and vessels that run with them. Name _____

C. Anterior and lateral leg (cadaver) COMPETENCY 127, LAB



Cadaver, lateral leg



Cadaver, anterior leg, deeper

1. Tibialis anterior
2. Fibularis longus
3. Superficial fibular nerve
4. Soleus
5. Extensor digitorum longus
6. Fibularis brevis

1. Anterior tibial artery
2. Extensor digitorum longus
3. Deep fibular nerve

FILL IN

Anterior leg muscles: dorsiflex, extend the toes

Fibularis longus and brevis: evert the foot

Common fibular nerve wraps the: neck of the fibula

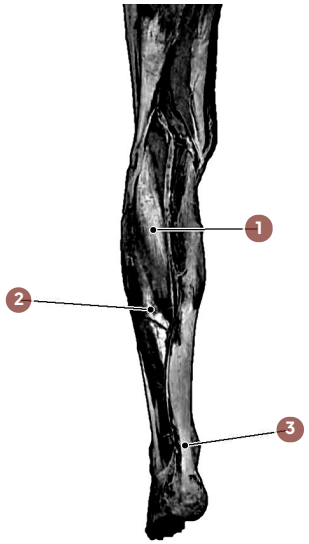
Deep branch runs the: anterior leg

Superficial branch runs the: lateral leg

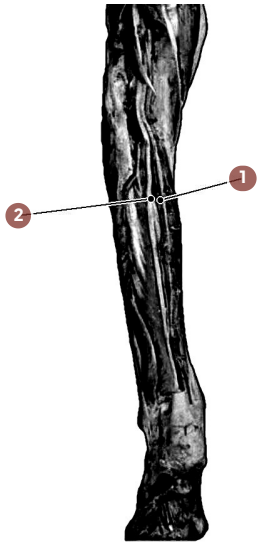
Injury at the fibular neck: foot drop

Anterior tibial artery becomes: dorsalis pedis

D. Posterior leg, and the veins (cadaver) COMPETENCY 127, LAB



Cadaver, posterior leg



Cadaver, deep posterior leg

1. Gastrocnemius
2. Soleus
3. Calcaneal tendon

1. Posterior tibial artery
2. Tibial nerve

ıemius and soleus: plantar flex the foot

re the: calcaneal (Achilles) tendon

ıe medial malleolus: posterior tibial artery, tibial nerve

ns run with the: arteries, same names

phenous vein: superficial, medial, in front of the medial malleolus, into the femoral vein

phenous vein: up the calf, into the popliteal vein

named at: inguinal ligament, adductor hiatus, below the knee

Extra notes and drawings

Extra notes and drawings

Lower Limb: Muscles, Nerves and Vessels

C. Anterior and lateral leg (cadaver) · Competency 127, lab · one card per numbered structure, same lines as the slide. Write the name, then fill each line as we go.

Name _____

1 Tibialis anterior

Def Most medial muscle of the anterior leg
Origin Lateral condyle and upper lateral tibia, interosseous membrane
Insertion Medial cuneiform and base of the first metatarsal
Action Dorsiflexes and inverts the foot
Nerve Deep fibular nerve (L4, L5)
Blood supply Anterior tibial artery
Clinical Shin splints; anterior compartment syndrome; paralysis gives foot drop

2 Fibularis longus

Def Superficial muscle of the lateral leg
Origin Head and upper two thirds of the fibula
Insertion Medial cuneiform and base of the first metatarsal, after crossing under the sole
Action Everts and plantar flexes the foot; supports the arches
Nerve Superficial fibular nerve (L5, S1)
Blood supply Fibular artery
Clinical Weak evertors mean repeated inversion ankle sprains

3 Superficial fibular nerve

Def Superficial branch of the common fibular nerve
Course Lateral compartment, then pierces the fascia to the top of the foot
Supplies Fibularis longus and brevis; skin of the lower lateral leg and top of the foot
Clinical Injury loses eversion; the foot tends to invert

4 Soleus

Def Flat calf muscle under gastrocnemius
Origin Soleal line of the tibia, upper fibula
Insertion Calcaneus by the calcaneal tendon
Action Plantar flexes the foot; holds you upright in standing and walking
Nerve Tibial nerve (S1, S2)
Blood supply Posterior tibial and fibular arteries
Clinical Deep veins inside it are a common site for DVT in a bedridden patient

5 Extensor digitorum longus

Def Anterior leg muscle lateral to tibialis anterior
Origin Lateral condyle of the tibia, upper fibula, interosseous membrane
Insertion Middle and distal phalanges of toes 2 to 5
Action Extends the toes, dorsiflexes the foot
Nerve Deep fibular nerve (L5, S1)
Blood supply Anterior tibial artery

6 Fibularis brevis

Def Lateral leg muscle under fibularis longus
Origin Lower two thirds of the fibula
Insertion Base of the fifth metatarsal
Action Everts the foot
Nerve Superficial fibular nerve (L5, S1)
Blood supply Fibular artery
Clinical A hard inversion can pull its tendon off the fifth metatarsal (avulsion fracture)

1 Anterior tibial artery

Def One of the two end branches of the popliteal artery
Course Passes forward through the interosseous membrane, down the anterior leg with the deep fibular nerve, onto the foot as the dorsalis pedis artery
Supplies Anterior compartment of the leg
Clinical Dorsalis pedis pulse on top of the foot; lost in peripheral arterial disease

2 Extensor digitorum longus

Def Anterior leg muscle lateral to tibialis anterior
Origin Lateral condyle of the tibia, upper fibula, interosseous membrane
Insertion Middle and distal phalanges of toes 2 to 5
Action Extends the toes, dorsiflexes the foot
Nerve Deep fibular nerve (L5, S1)
Blood supply Anterior tibial artery

3 Deep fibular nerve

Def Deep branch of the common fibular nerve
Course Runs with the anterior tibial artery in the anterior compartment
Supplies Tibialis anterior, extensor digitorum longus, extensor hallucis longus, fibularis tertius; skin between toes 1 and 2
Clinical Damage gives foot drop and a high-stepping gait

Lower Limb: Muscles, Nerves and Vessels

D. Posterior leg, and the veins (cadaver) · Competency 127, lab · one card per numbered structure, same lines as the slide. Write the name, then fill each line as we go.

Name _____

1 Gastrocnemius

Def Superficial calf muscle with two heads
Origin Medial and lateral condyles of the femur
Insertion Calcaneus by the calcaneal tendon
Action Plantar flexes the foot, flexes the knee
Nerve Tibial nerve (S1, S2)
Blood supply Sural branches of the popliteal artery
Borders Its two heads form the lower edges of the popliteal fossa
Clinical Tennis leg: tear of the medial head; calf pump failure contributes to DVT

2 Soleus

Def Flat calf muscle under gastrocnemius
Origin Soleal line of the tibia, upper fibula
Insertion Calcaneus by the calcaneal tendon
Action Plantar flexes the foot; holds you upright in standing and walking
Nerve Tibial nerve (S1, S2)
Blood supply Posterior tibial and fibular arteries
Clinical Deep veins inside it are a common site for DVT in a bedridden patient

3 Calcaneal tendon

Def The Achilles tendon, thickest tendon in the body
Contributing structures Gastrocnemius and soleus share it
Insertion Posterior calcaneus
Function Transfers the pull of both muscles to plantar flex the foot
Clinical Rupture gives a sudden pop and a positive Thompson test (squeeze the calf, no plantar flexion)

1 Posterior tibial artery

Def Larger end branch of the popliteal artery
Course Down the back of the leg deep to the calf muscles, beside the tibial nerve, behind the medial malleolus
Supplies Posterior compartment of the leg and, through the plantar arteries, the sole
Clinical Pulse felt behind the medial malleolus

2 Tibial nerve

Def Larger branch of the sciatic nerve
Course Through the popliteal fossa, down the back of the leg beside the posterior tibial artery, behind the medial malleolus, into the sole
Supplies Posterior leg muscles and the sole of the foot
Clinical Tarsal tunnel syndrome behind the medial malleolus; injury loses plantar flexion and the ankle reflex

Lymphatic System

Competencies 105 to 108: the route lymph takes, vessel and node structure, primary and secondary organs, disorders.

Name _____

A. The route lymph takes (draw it) and the spleen (cadaver) COMPETENCY 105, 107

ONE-WAY ROUTE, IN ORDER

1. lymphatic capillaries (blind ends in the tissue)
2. collecting vessels, with valves
3. lymph nodes
4. lymphatic trunks
5. right lymphatic duct or thoracic duct
6. subclavian veins

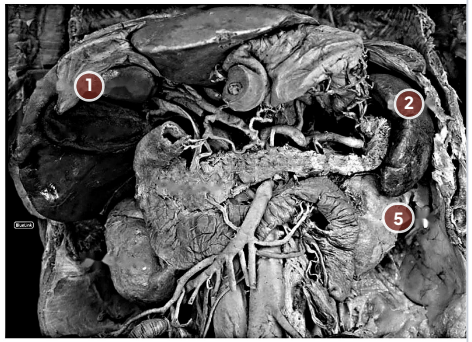
FILL IN

Lymph is fluid the capillaries left in the tissue, picked up by lymphatic vessels

Right lymphatic duct drains the right arm, right head and chest (upper right quarter)

Thoracic duct drains everything else; starts at the cisterna chyli

Lymph moves with no pump by skeletal muscle, breathing, valves



Cadaver, upper abdomen: find the spleen

1. Right lobe of liver
2. Spleen
5. Left kidney

SPLEEN

Sits upper left, behind the stomach, under ribs 9 to 11

White pulp: lymphocytes, immune

Red pulp: blood filtering, old red cells removed

Extra notes and drawings

B. Lymphatic capillary, collecting vessel, lymph node (draw it) COMPETENCY 106

LYMPH NODE, OUTSIDE IN

Capsule: dense connective tissue

Cortex: lymphatic nodules, lymphocytes

Medulla: medullary cords and sinuses

Afferent vessels enter the convex side, many

Efferent vessels leave the hilum, few

Hilum: artery in, vein and efferent vessels out

CAPILLARY AND VESSEL

Lymphatic capillary wall: overlapping endothelial flaps, one-way minivalves

Collecting vessels have valves, so flow is one way

Lacteals are lymphatic capillaries in the villi of the small intestine

DRAW: ONE LYMPH NODE WITH CAPSULE, CORTEX, MEDULLA, AFFERENT VESSELS ENTERING AND EFFERENT VESSELS LEAVING THE HILUM

Extra notes and drawings

Lymphatic System

A. The route lymph takes (draw it) and the spleen (cadaver) · Competency 105, 107 · one card per numbered structure, same lines as the slide. Write the name, then fill each line as we go.

Name _____

1 Right lobe of liver

Def Largest lobe of the liver

Location Right of the falciform ligament, under the right dome of the diaphragm, behind ribs 7 to 11

Blood supply Right branches of the hepatic artery proper and hepatic portal vein

Note The gallbladder sits in a hollow on its underside

2 Spleen

Def Largest lymphatic organ

Location Upper left abdomen, behind the stomach, under ribs 9 to 11

Borders Diaphragm above and behind; stomach in front; left kidney behind and below; splenic flexure below; the superior border is notched

Parts White pulp (lymphocytes around arterioles) and red pulp (blood-filled sinuses)

Blood supply Splenic artery (celiac trunk); splenic vein to the hepatic portal vein

Function Fights blood-borne infection, removes old red cells, stores platelets

Clinical Ruptures with a blow to the left lower ribs; a person without a spleen needs vaccines against encapsulated bacteria

5 Left kidney

Def Retroperitoneal organ against the posterior abdominal wall

Location T12 to L3, beside the vertebral column; sits a little higher than the right kidney

Borders Spleen, stomach, pancreas, and descending colon in front; diaphragm and quadratus lumborum behind

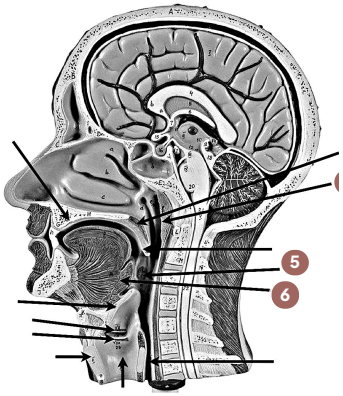
Blood supply Left renal artery from the aorta; left renal vein crosses in front of the aorta

Lymphatic System

Competencies 105 to 108: the route lymph takes, vessel and node structure, primary and secondary organs, disorders.

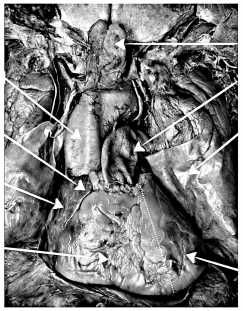
Name _____

C. Primary and secondary organs COMPETENCY 107



Model, midsagittal head: find the three tonsils

3 Pharyngeal tonsil
5 Palatine tonsil
6 Lingual tonsil



adaver, chest opened: find the thymus

Thymus

FILL IN

Primary organs: red bone marrow, thymus (lymphocytes made, mature)

Secondary organs: nodes, spleen, tonsils, MALT (where they work)

Thymus: mediastinum, behind the sternum; big in children

Pharyngeal tonsil: nasopharynx

Palatine tonsils: oropharynx

Lingual tonsil: root of the tongue

MALT: Peyer patches, appendix

Extra notes and drawings

D. Disorders: which structure does each one affect?

COMPETENCY 108

DISORDER: STRUCTURE, WHAT CHANGES

Edema: tissue fluid builds up, any cause

Lymphedema: lymph vessels blocked or removed, limb swells

Lymphadenopathy: lymph nodes enlarged, infection or cancer

Lymphoma: cancer of the lymphocytes in lymphatic tissue

Splenomegaly: spleen enlarged; rupture risk in contact sports

Tonsillitis: palatine tonsils inflamed

DRAW: THE BODY IN OUTLINE, THE THORACIC DUCT AND RIGHT LYMPHATIC DUCT, AND SHADE THE QUARTER THE RIGHT DUCT DRAINS

Extra notes and drawings

Lymphatic System

C. Primary and secondary organs · Competency 107 · one card per numbered structure, same lines as the slide. Write the name, then fill each line as we go.

Name _____

3 Pharyngeal tonsil

Def Single mass of lymphatic tissue

Location Back wall of the nasopharynx

Function Traps pathogens entering through the nose

Clinical Enlarged adenoids block the nose and the auditory tube, so children mouth-breathe and get ear infections

5 Palatine tonsil

Def Paired lymphatic tissue

Location Side walls of the oropharynx, between the palatoglossal and palatopharyngeal arches

Blood supply Tonsillar branch of the facial artery

Function Traps pathogens entering through the mouth

Clinical Tonsillitis; tonsillectomy bleeding comes from the tonsillar artery

6 Lingual tonsil

Def Lymphatic tissue on the root (back) of the tongue

Location Oropharynx, below the palatine tonsils

1 Thymus

Def Primary lymphatic organ

Location Superior mediastinum, behind the manubrium, in front of the great vessels

Borders Sternum in front; pericardium and aortic arch behind; pleura on each side

Blood supply Internal thoracic and inferior thyroid arteries

Function T lymphocytes mature here

Clinical Large in children and shrinks after puberty; thymoma is linked to myasthenia gravis